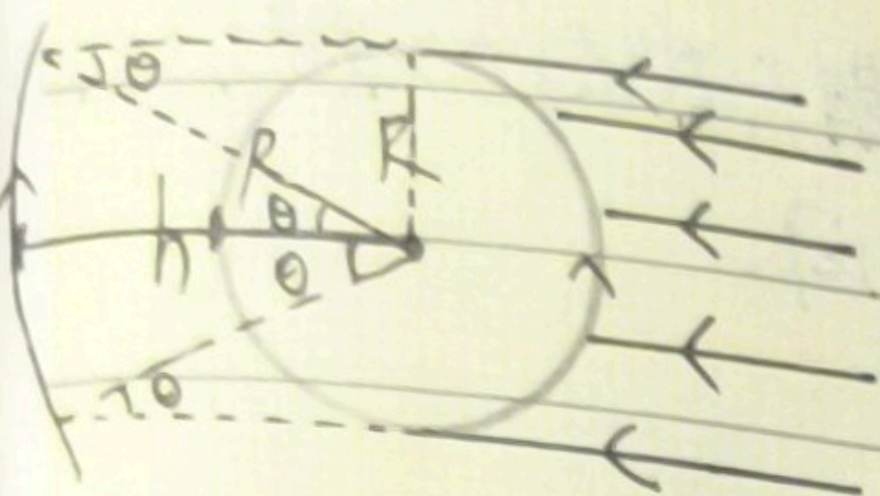


3. 飞船 $h=340\text{ km}$ 绕地球一周 90 min $R_{\text{地}}=6400\text{ km}$. 飞船沿赤道平面自东向西飞行.



飞行看到西边日出
若太阳直射赤道, 飞行员每天看 n 次日落 $n = \frac{24\text{ h}}{1.5\text{ h}} = 16\text{ 次}$

飞船

一周中飞船看不见太阳的时间有多长? $\cos 18^\circ = 0.95$ $\tan 18^\circ = 0.32$ $T = 90\text{ min} = 1.5\text{ h}$

$$\sin \theta = \frac{R}{R+h} = 0.95$$

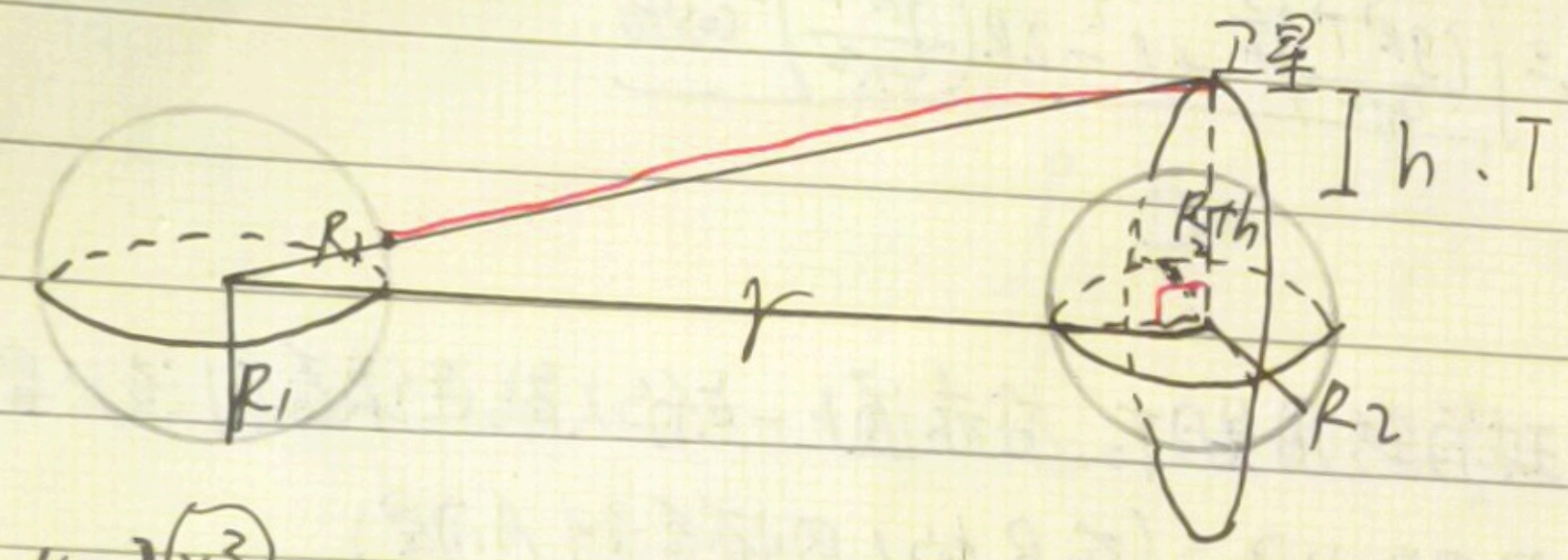
$$\therefore \theta = 72^\circ$$

$$t = \frac{2\theta}{360^\circ} T = \frac{2}{5} T$$

$$T = 24\text{ h} \quad \therefore t = 36\text{ min}$$

$$T = 90\text{ min}$$

4.



月绕地周期 T_0 半径为 r

$$M = \frac{4\pi^2 r^3}{G T^2}$$

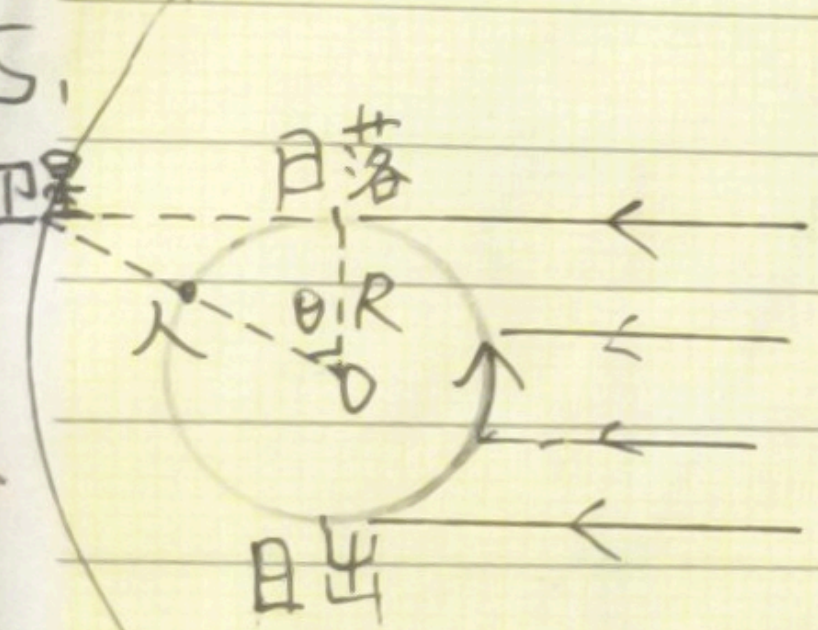
$$\frac{M_A}{M_{\text{地}}} = \frac{(h+R_2)^3 T_0^2}{r^3 T^2}$$

卫星向地球发送照片的时间? 光速 c

$$d = \sqrt{r^2 + (R_2+h)^2} - R_1$$

$$t = \frac{d}{c} = \frac{\sqrt{r^2 + (R_2+h)^2} - R_1}{c}$$

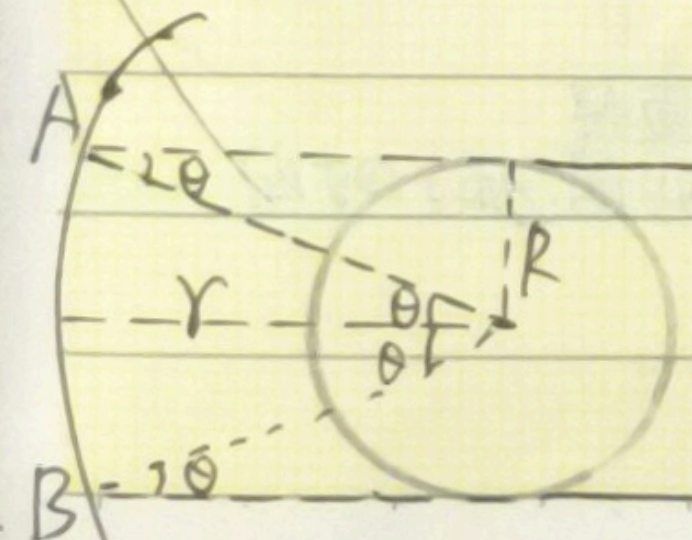
5.



赤道上方飞行的
人在地上看到明亮的卫星, 日落 2 h 以后赤道附近的人们能看见它, 求它最低高度.

$$\theta = \frac{360^\circ}{12} = 30^\circ \quad h = \frac{24}{2} = 12$$

$$\cos \theta = \frac{R}{R+h} \quad \therefore h = \left(\frac{2}{3} - 1 \right) R$$



看 某地球同步卫星正下方的地球表面有一观察者, 他用天文望远镜观察被太阳光照射的此卫星. 太阳直射赤道(看为) 日落有 12 h 看不见卫星, 有几个小时看不见卫星. (地表重力加速度 g , 地球自转周期 T .)

$$\sin \theta = \frac{R}{R+h}$$

$$\theta = \arcsin \frac{R}{R+h} \quad t = \frac{2\theta}{360^\circ} T$$

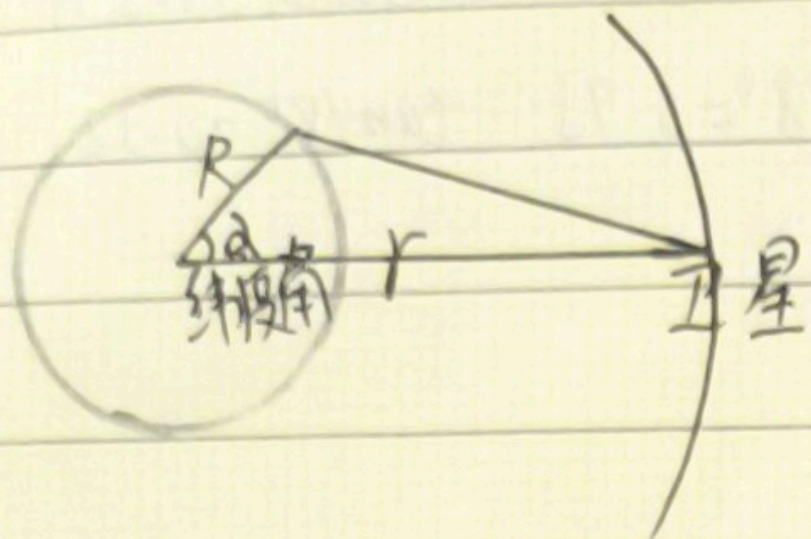
对同步卫星

$$\frac{GMm}{r^2} = \frac{m 4\pi^2}{T^2} r$$

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}} = \sqrt[3]{\frac{gR^2 T^2}{4\pi^2}}$$

$$t = \frac{\arcsin \frac{R}{R+h}}{\pi} T$$

6. 我国同步卫星、定点位置与东经98°经线在同一平面内。把甘肃省嘉峪关外的经纬度和纬度近似为东经98°和北纬 $\lambda=40^\circ$ 。地球半径 R ，自转周期 T ，地表重力加速度 g ，光速 c ，求该同步卫星发出微波信号传到嘉峪关所需时间。



$$l = \sqrt{r^2 + R^2 - 2rR \cos \alpha}$$

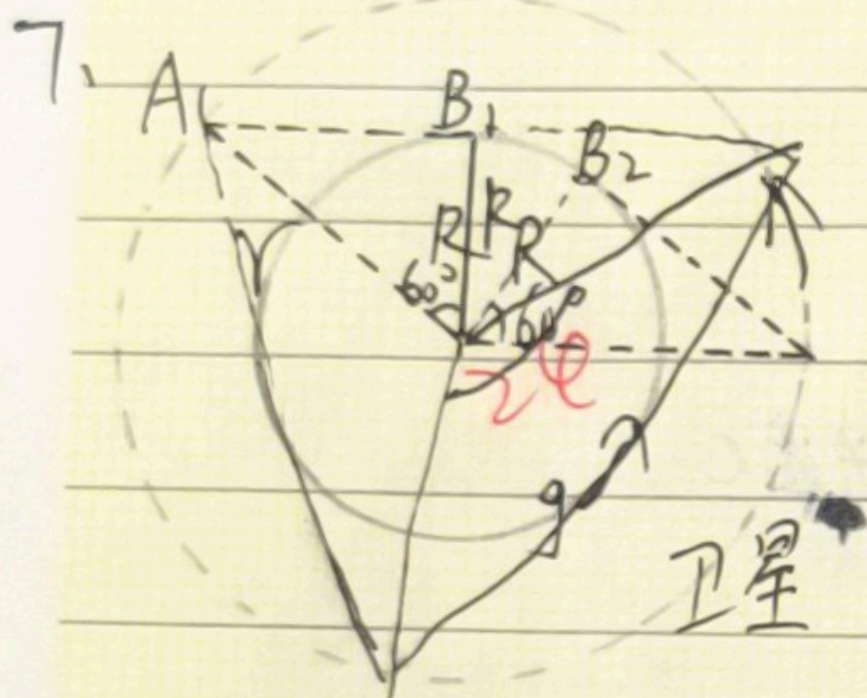
由 $\frac{GMm}{r^2} = m \frac{4\pi^2}{T^2} r$ 和 $gR^2 = GM$

$$\therefore r = \sqrt[3]{\frac{gR^2 T^2}{4\pi^2}}$$

$$t = \frac{l}{c}$$

解出 $t = \frac{\sqrt{\left(\frac{gR^2 T^2}{4\pi^2}\right)^2 + R^2 - 2R\left(\frac{gR^2 T^2}{4\pi^2}\right)^{\frac{1}{3}} \cos 40^\circ}}{c}$

C



地球自转周期 T_0 ，在赤道上一点的人能连续看到该卫星的时间是多少？（在 B_1 的人刚好看到A卫星）

$$\omega t - \omega_0 t = \frac{2\pi}{3}$$

$$\left(\frac{2\pi}{T} - \frac{2\pi}{T_0}\right)t = \frac{2\pi}{3}$$

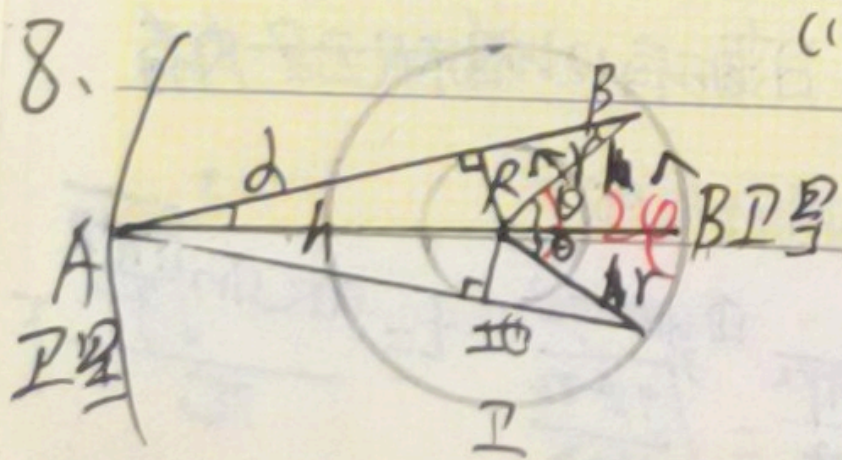
$$\begin{cases} R+r = 2R \\ \frac{GMm}{r^2} = m \frac{4\pi^2}{T^2} r \end{cases}$$

$$\therefore t = \frac{4\pi \sqrt{\frac{2R}{g}} T}{3(T_0 - 4\pi \sqrt{\frac{2R}{g}})}$$

$$T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{8R^3}{gR^2}} = 4\pi \sqrt{\frac{2R}{g}}$$

$$= 2\pi \sqrt{\frac{R}{g}} = 2\pi \sqrt{\frac{2R}{g}}$$

自转方向与地球自转方向一样
地球自转周期 T_0 和 R ，(1)求 B 的周期(2)A、B不能通讯的时间



(1)

$$\text{由开三} \quad \frac{r^3}{T^2} = \frac{R^3}{T_0^2}$$

$$T_B = T \sqrt{\frac{r^3}{R^3}}$$

sin

$$(2) \omega_B t - \omega t = 2\pi$$

$$\sin \beta = \frac{R}{r}$$

$$\sin \alpha = \frac{R}{r}$$

$$\theta = \alpha + \beta$$

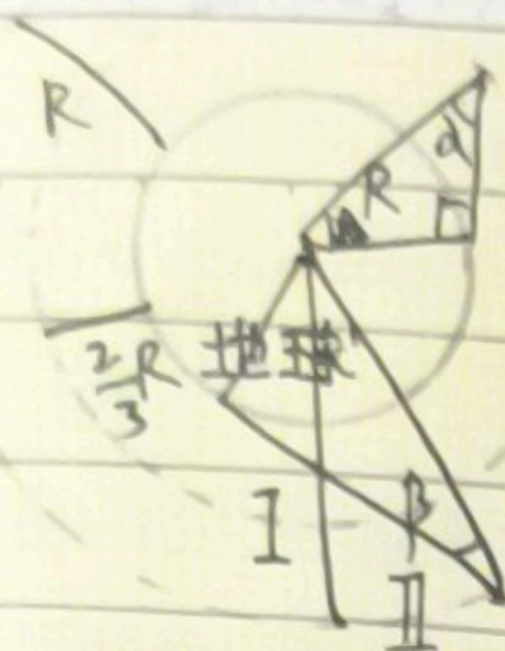
$$\therefore t =$$

作切线类

NO. 1.429 g.t.

No.

AB两卫星某时刻相距最近,若该时刻开始至两卫星再次相距最近的时间为T.在这段时间T内两卫星可直接利用微波通信的时间为?



$$\omega_A t - \omega_B t = 2\pi \quad \text{①} \quad \omega_A T - \omega_B T = 2\pi \quad \text{②} \quad \frac{\text{①}}{\text{②}} \Rightarrow \frac{t}{T} = \frac{67}{180}$$

$$\sin \alpha = \frac{R}{R + \frac{2R}{3}} = \frac{3}{5} \quad \alpha = 37^\circ$$

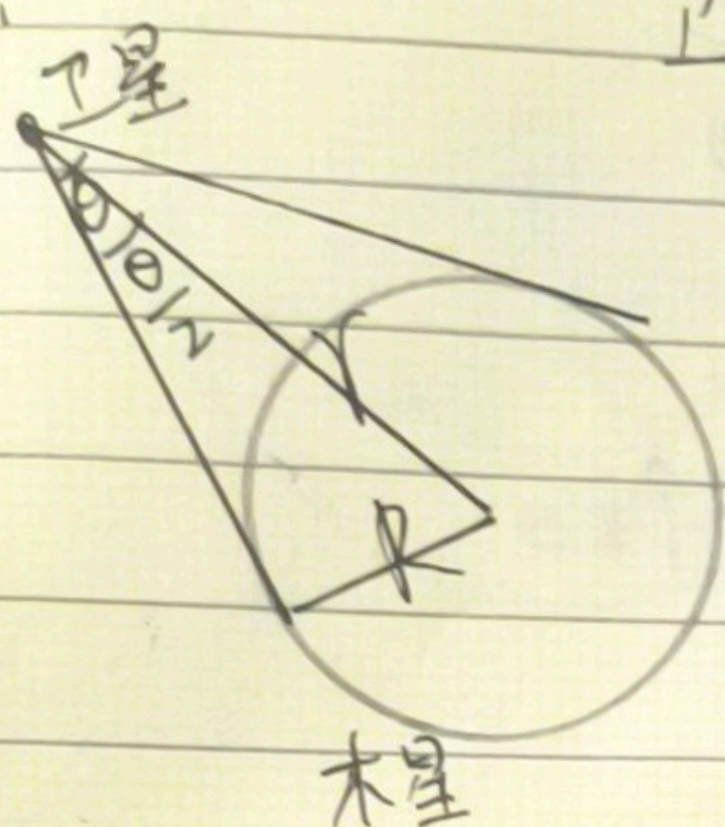
$$\sin \beta = \frac{R}{R + R} = \frac{1}{2} \quad \beta = 30^\circ$$

时间t应大于 $\frac{67}{180} T$

(不能等于)

9.

卫星t绕木星N圈速率为V.求轨道半径和木星第一宇宙速度



$$\text{由 } t = \frac{N \cdot 2\pi r}{V}$$

$$\therefore r = \frac{Vt}{2\pi N}$$

$$\frac{GMm}{r^2} = \frac{mV^2}{r}$$

$$V_1 = \sqrt{\frac{GM}{R}}$$

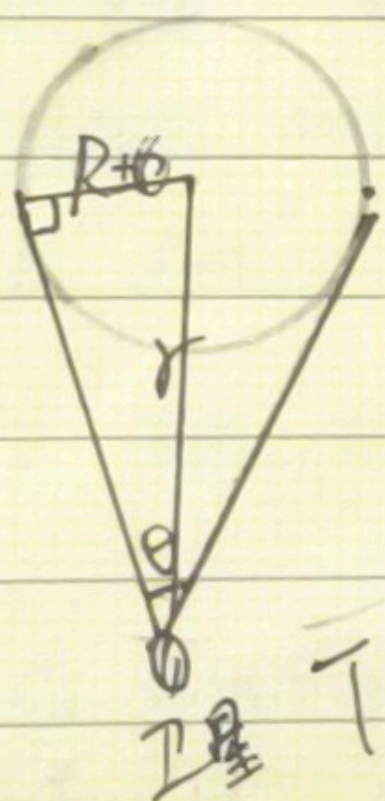
$$\left\{ \begin{array}{l} \text{卫星} \quad \frac{GMm'}{r^2} = \frac{m'V^2}{r} \\ R = r \sin \frac{\theta}{2} \\ GM = V_1^2 R \\ \therefore V_1 = \frac{V}{\sqrt{\sin \frac{\theta}{2}}} \end{array} \right.$$

$$R = r \sin \frac{\theta}{2}$$

$$GM = V_1^2 R$$

$$\therefore V_1 = \frac{V}{\sqrt{\sin \frac{\theta}{2}}}$$

求地球密度



$$\rho = \frac{3\pi}{GT^2}$$

$$\left(\frac{R+\delta}{r} \right)^3 = \left(\frac{T+\delta}{T} \right)^2$$

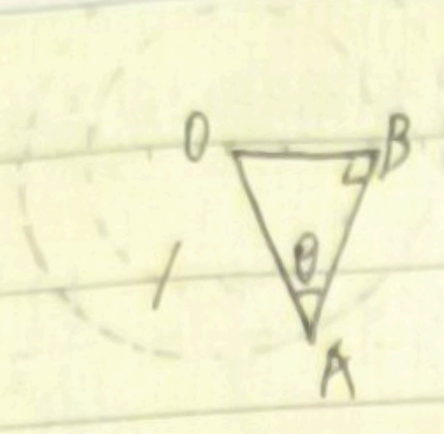
$$\left(\frac{R+\delta}{r} \right)^3 = \frac{T^2}{T+\delta}$$

$$\frac{R+\delta}{r} = \sin \frac{\theta}{2}$$

$$\therefore \left(\sin \frac{\theta}{2} \right)^{\frac{3}{2}} T = T_{\pm \theta}$$

$$\rho = \frac{3\pi}{G \sin^{\frac{3}{2}} \frac{\theta}{2} T^2}$$

求卫星 $V_A = V_B$ A、B 与 A、O 间连线间夹角最大为 θ



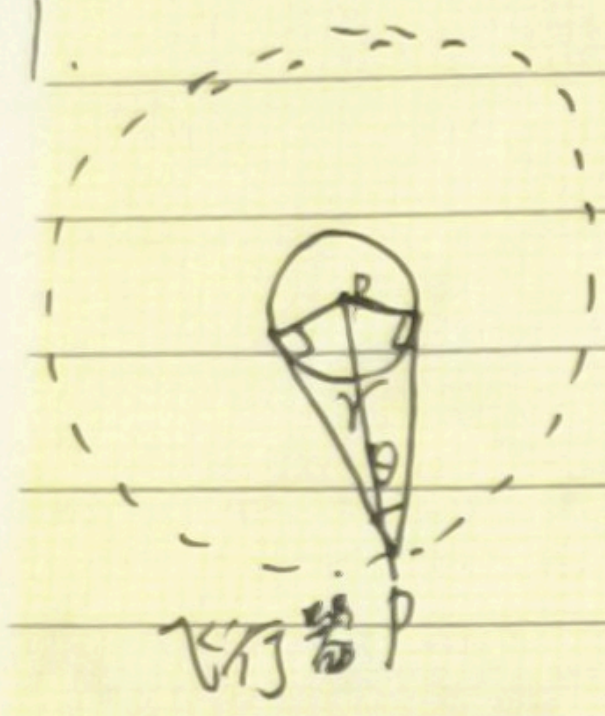
$$V = \sqrt{\frac{GM}{r}} \quad T = \sqrt{\frac{4\pi^2 r^3}{GM}} \quad \omega = \sqrt{\frac{GM}{r^3}}$$

$$V \propto \frac{1}{\sqrt{r}} \quad \sin \theta = \frac{r_B}{r_A}$$

$$\frac{V_A}{V_B} = \sqrt{\frac{r_B}{r_A}} = \sqrt{\sin \theta}$$

$$\frac{T_A}{T_B} \propto \sqrt{\frac{R_B^3}{R_A^3}} = \sqrt{\sin^3 \theta} \quad \frac{\omega_A}{\omega_B} \propto \frac{\sqrt{R_B^3}}{\sqrt{R_A^3}} = \sqrt{\sin^3 \theta}$$

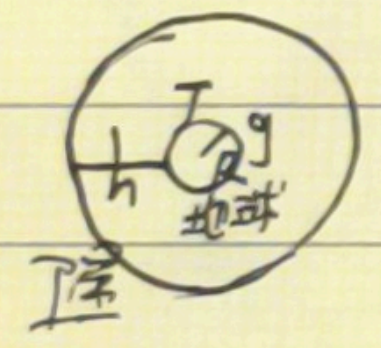
11.



$$R = r \sin \frac{\theta}{2} \quad T = \sqrt{\frac{4\pi^2 r^3}{GM}} \propto r^{\frac{3}{2}} \quad r \uparrow T \uparrow$$

$$\rho = \frac{3\pi}{GT^2 (\sin \frac{\theta}{2})^3} \quad \text{算密度要周期和张角}$$

12.

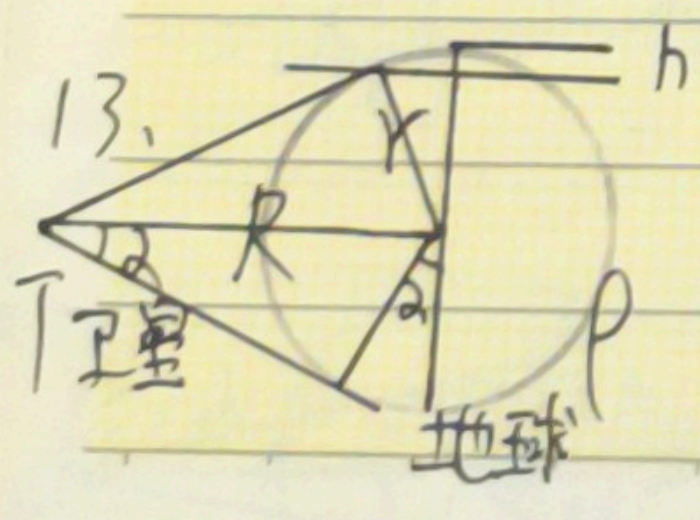


卫星速度 和 1 天内将地球上赤道各处在日照条件下的情况拍下来，卫星在每次通过赤道上空时，卫星上的摄像机应拍摄地球上赤道的弧长 s 是多少？

$$\begin{cases} \text{由 } \frac{GMm}{(R+h)^2} = \frac{mV^2}{(R+h)} \\ m_0 g = \frac{GMm_0}{R^2} \end{cases} \quad \begin{cases} \frac{GMm}{(R+h)^2} = m \frac{4\pi^2}{T^2} (R+h) \\ T' = \frac{2\pi}{R} \sqrt{\frac{(R+h)^3}{g}} \end{cases}$$

$$V = \frac{R}{R+h} \sqrt{g(R+h)} \quad N = \frac{T}{T'}$$

$$s = \frac{2\pi R}{N} = \frac{4\pi^2}{T^2} \sqrt{\frac{(R+h)^3}{g}}$$



求摄像机所能拍摄总面积与地球表面积之比。 $V_{\text{球}} = \frac{4}{3}\pi R^3$ $S_{\text{球}} = 4\pi R^2$

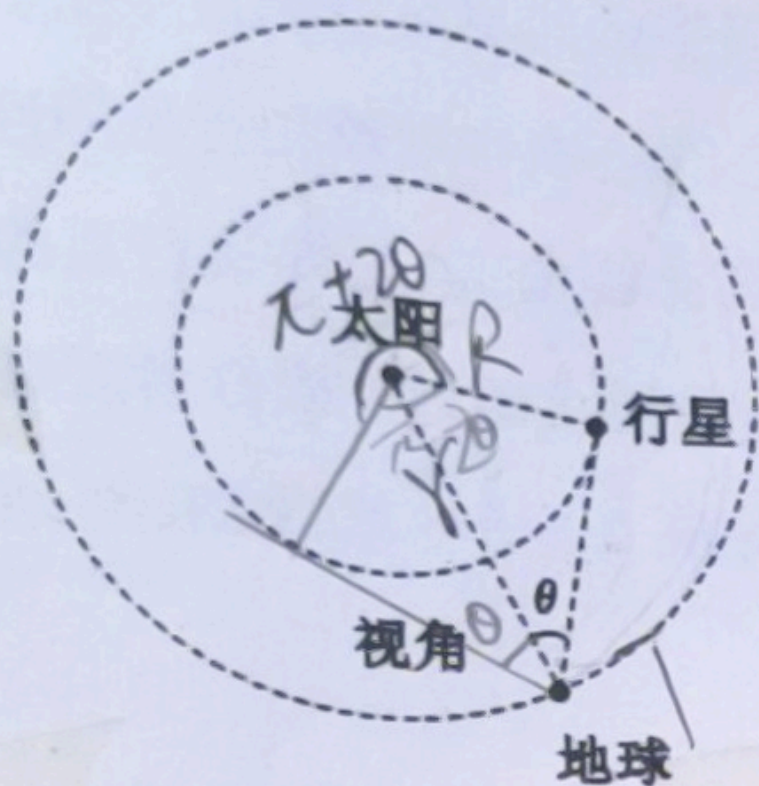
$$\cos \alpha = \frac{\sqrt{R^2 - r^2}}{R} \quad h = r(1 - \cos \alpha)$$

$$\eta = \frac{S - S_0}{S} = \frac{4\pi R^2 - 2 \cdot 2\pi R h}{4\pi R^2} = \frac{R-h}{R} = 1 - (1 - \cos \alpha) = \cos \alpha$$

$$\cos \alpha = \sqrt{1 - \left(\frac{r}{R}\right)^2} = \sqrt{1 - \left(\frac{3\pi}{GPT^2}\right)^{\frac{2}{3}}}$$

B7、地球和某行星在同一轨道平面内同向绕太阳做匀速圆周运动。地球的轨道半径为 r ，运转周期为 T 。地球和太阳中心的连线与地球和行星的连线所夹的角叫地球对该行星的观察视角（简称视角）。当行星处于最大视角 θ 处时（如图所示），是地球上天文爱好者观察该行星的最佳时期。

- 求：(1) 该行星的公转周期 T_2 ；
(2) 若某时刻该行星正处于最佳观察期，则该行星下一次处于最佳观察期需要多长时间 t ？



(1)

再次相遇

$$t = \frac{T_1 T_2}{T_1 - T_2}$$

但添加手数 $\frac{\pi + 2\theta}{2\pi}$
外用角 2π

$$t = \frac{(\pi + 2\theta) \sqrt{\sin^3 \theta}}{2\pi (1 - \sqrt{\sin \theta})} T$$

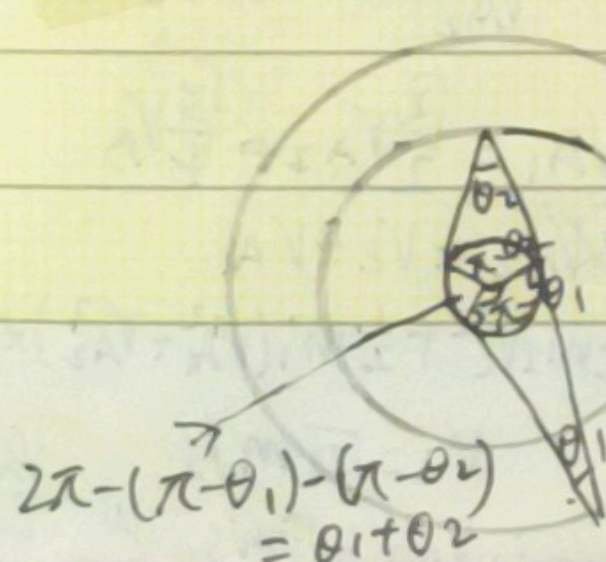
B8、我国的“天链一号”卫星是地球同步卫星，可为中低轨道卫星提供数据通讯，如图所示为“天链一号”卫星 a、赤道平面内的低轨道卫星 b、地球的位置关系示意图，O 为地心，地球相对卫星 a、b 的张角分别为 θ_1 弧度和 θ_2 弧度 (θ_2 图中未标出)，卫星 a 的轨道半径是 b 的 4 倍，已知卫星 a、b 绕地球同向运行，卫星 a 的周期为 T ，在运行过程中由于地球的遮挡，卫星 b 会进入卫星 a 通讯的盲区，卫星间的通讯信号视为沿直线传播，信号传输时间可忽略。下列分析正确的是

A. 卫星 a、b 的速度之比为 2 : 1

B. 卫星 b 的周期为 $\frac{T}{8}$

C. 卫星 b 每次在盲区运行的时间 $\frac{\theta_1 + \theta_2}{14\pi} T$

D. $\frac{\theta_1 + \theta_2}{16\pi} T$



外用角 $\theta_1 + \theta_2$
 $\frac{\theta_1 + \theta_2}{2\pi}$

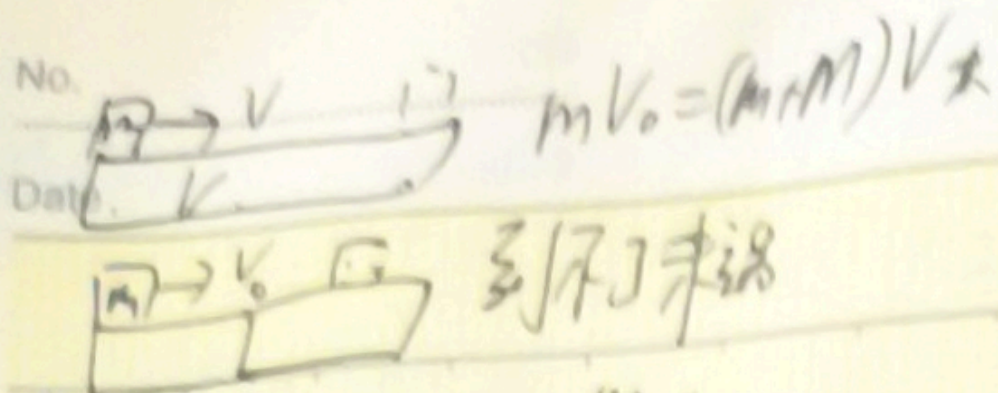
$$t = \frac{\theta_1 + \theta_2}{2\pi} \frac{T}{7} = \frac{\theta_1 + \theta_2}{14\pi} T$$

$$\frac{T_a}{T_b} = \left(\frac{r_a}{r_b}\right)^{\frac{3}{2}} = 8$$

$$t_{\text{盲区}} = \frac{T \frac{7}{8}}{T - \frac{T}{8}} = \frac{T}{7}$$

No.

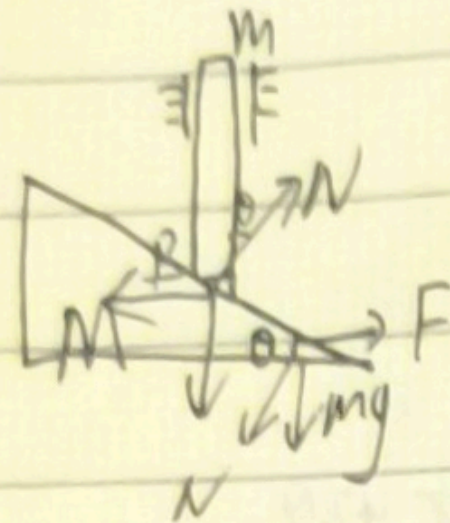
Date.



$$mV_0 = (m + \frac{M}{2})V_2 + \frac{M}{2}V_1$$

$$(m + M)V_0 = (m + \frac{M}{2})V_2 + \frac{M}{2}V_1$$

1.



$$\begin{cases} N \cos \theta = mg \\ F = N \sin \theta \\ N = N' \\ F = mg \tan \theta \end{cases}$$

① 对斜面施加 F 使之平衡求 F . ② 用 F 使斜面缓慢右移, m 上升 h , 求斜面位移 S , 及 F 做的功.

$$V_{\text{杆}} = V_{\text{斜}} = 0$$

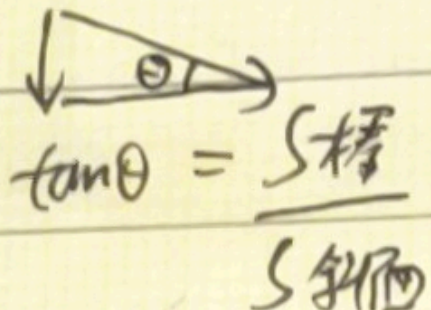
$$W = mgh, S = \frac{h}{\tan \theta}$$

③ 初始时 B 点高 h_2 . 由静止释放 m, M . 求 m 着地时 M 速度

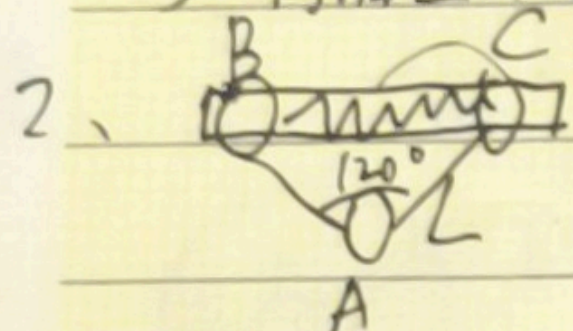
$$mgh_2 = \frac{1}{2}mV_1^2 + \frac{1}{2}MV_2^2$$

$$V_1 = V_2 \tan \theta$$

$$V_2 = \sqrt{\frac{2mgh_2}{m \tan^2 \theta + M}}$$



3. 简谐运动对称性 $\angle A$ 从 $120^\circ \rightarrow 60^\circ$ 过程



A 刚开 $a_A = g$ 向下.

故 A 在最低点时 $a_A = g$ 向上

$$2T \cos 30^\circ - mg = ma \quad T = \frac{2\sqrt{3}}{3}mg$$

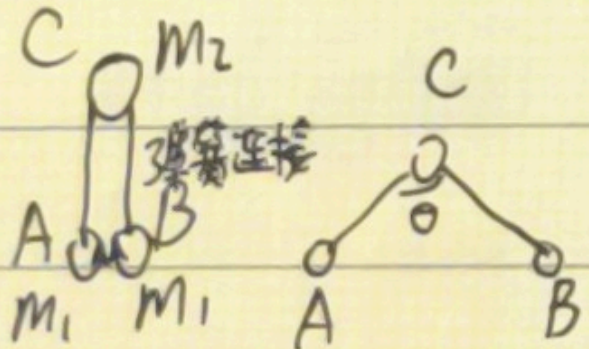
$$\therefore N_{\text{杆}A} = 2mg$$

$$E_p = mg(h_1 - h_2) = \frac{\sqrt{3}-1}{2}mgL$$

$$W_{\text{轻绳}B} = \frac{\sqrt{3}-1}{4}mgL \quad W_{\text{轻绳}A} = \frac{\sqrt{3}-1}{4}mgL$$

P 绳 B 先上后下 V 为 0 有 0

3.

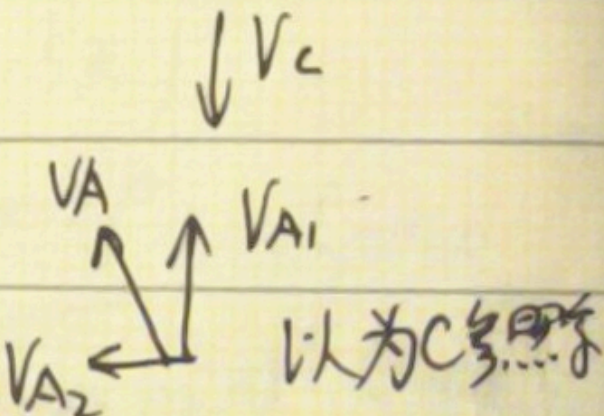


释放后 球与弹簧分离时杆弹力为 10N

① 求 E_p .

② 若 C 球不固定, 求 $V_{C \max}$

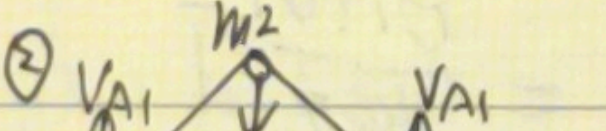
③ 若 C 球不固定求 $\theta = 120^\circ$ 第一次时, A 绕 C 的 W



$$\textcircled{1} E_p = 2 \cdot \frac{1}{2}mv^2$$

$$F = \frac{mv^2}{L}$$

$$\text{故 } E_p = 10J$$



$$m_1V_{A1} + m_1V_{A1} = m_2V_C$$

$$\frac{1}{2}m_1V_{A1}^2 + \frac{1}{2}m_1V_{A1}^2 + \frac{1}{2}m_2V_C^2 = E_p$$

$$V_{A1} = V_C \text{ 时, } V_{A2} = 0 \text{ 时 } V_{C \max}$$

$$\textcircled{3} V_C = V_{A1} = \frac{\sqrt{3}}{2}V_{A2} = \frac{\sqrt{3}}{4}V_A$$

$$\sqrt{3}V_{A2} = V_C + V_{A1}$$

$$\frac{1}{2}m_2V_C^2 + \frac{1}{2}m_1(V_{A1}^2 + V_{A2}^2) \times 2 =$$

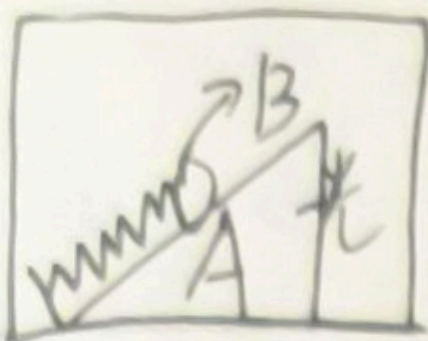
$$V_C = \sqrt{15}m/s$$

$$\therefore V_A = 4\sqrt{5}m/s \quad W = \frac{V_A^2}{L} = 40J$$

$$\text{故 } E_p = \frac{1}{2}m_2V_C^2 + \frac{1}{2}m_1V_{A1}^2 \times 2$$

$$V_A = V_C$$

$$\text{故 } V_C = 5m/s$$



$A: B$ 动能增加量 = 斜面支持力和弹簧弹力做功之和
 $B: B$ 机械能增加量 = 斜面支持力和弹簧弹力做功之和
 $C: B$ 和弹组机械能 = 斜面支持力和弹簧弹力做功之和

NO. 14
 $\rho = 1.429 g$
 $W_B = W_A + W_{\text{弹}} + W_N$

绳杆连接体

- ① 牛顿二律 $\Sigma F = \Sigma (ma)$
- ② 系统动能定理 $\Sigma I_{\text{外}} = \Sigma (P_{\text{外}})$
- ③ 系统动能定理 $\Sigma W_{\text{外}} + \Sigma W_{\text{内}} = \Sigma (E_k)$

区别 ① 不考虑内力

② 要考虑内力 不能忽略电场力

机守 C 能守 C 动能定理

内力 $\times$ 相对距离

- 安培力
- 库仑力
- 弹簧

绳杆动能定理

- 1: 物1 动能定理
- 2: 物2 动能定理
- 关联速度 V_1 与 V_2 关系 (沿绳/杆速度都相等)

恒定电流

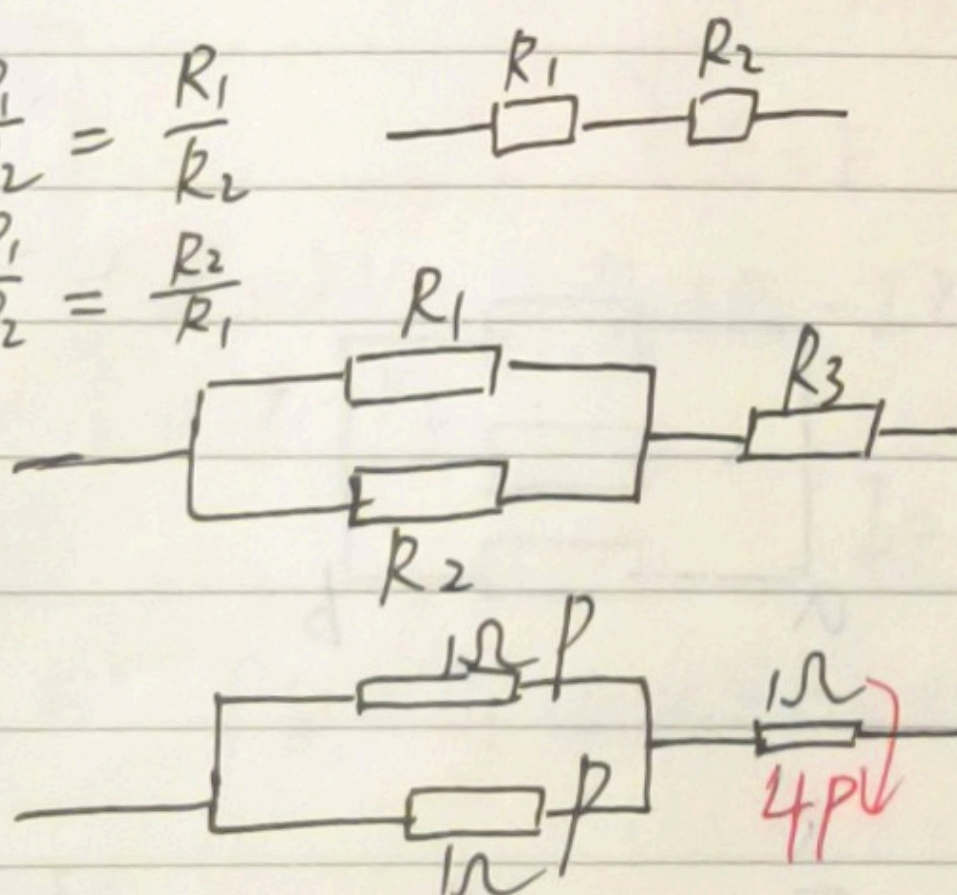
功率 $\begin{cases} P = UI : \text{任意用电器} \\ P = I^2 R : \text{任意用电器热功率 电阻为热源} \end{cases}$

$U = IR$ 纯电阻

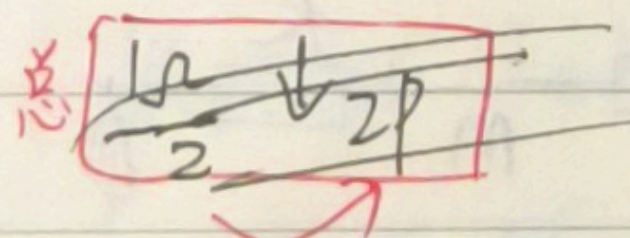
功率分配

串 $\frac{P_1}{P_2} = \frac{R_1}{R_2}$
 并 $\frac{P_1}{P_2} = \frac{R_2}{R_1}$

$P_{\text{总}} = P_1 + P_2$
 $P_{\text{总}} = P_1 + P_2$



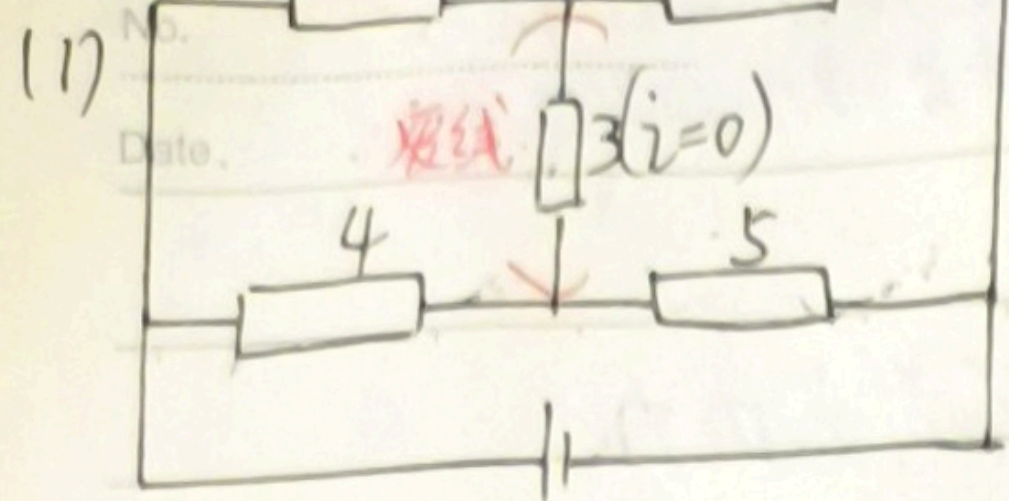
$P_{\text{总}} = 6P$
 $1:1:4$



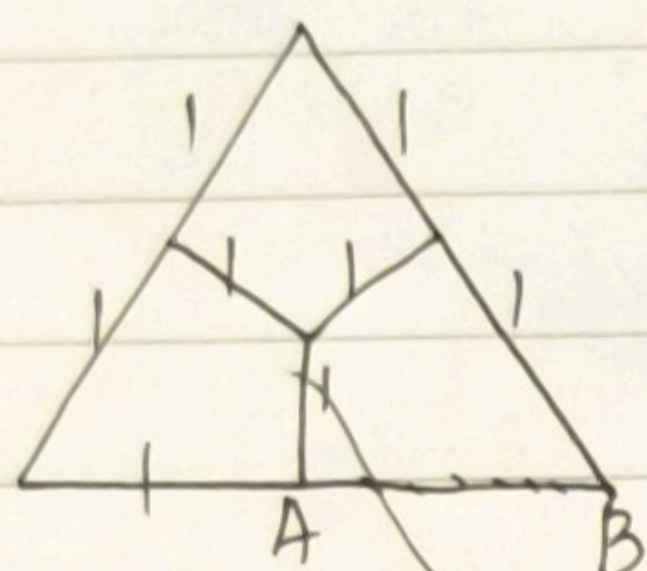
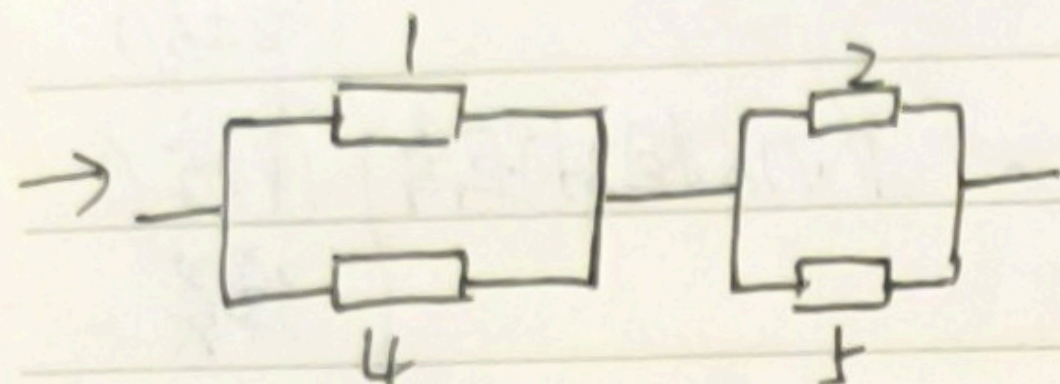
总体上看 $\frac{1}{2}\Omega \text{ 与 } 2P \Rightarrow 1\Omega \text{ 与 } 4P$

电路化简

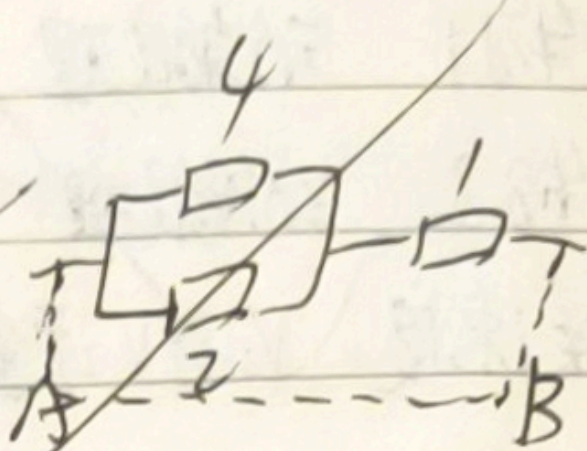
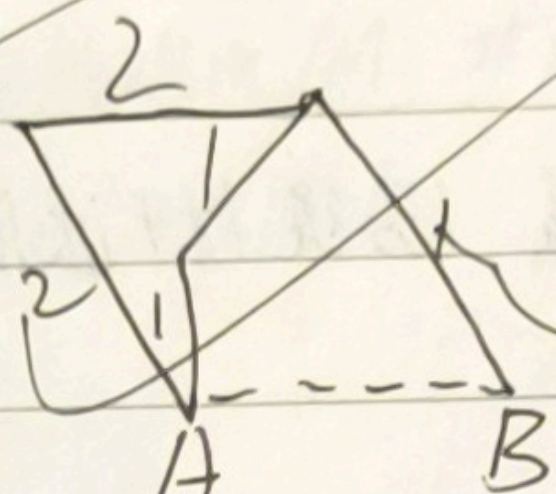
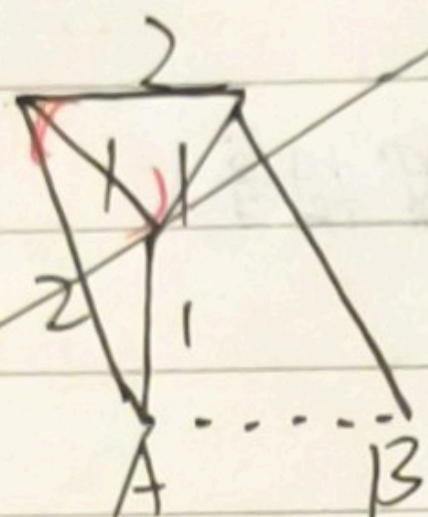
该图无法化简为串并联



(2) 惠斯通电桥 $\frac{R_1}{R_2} = \frac{R_4}{R_5}$ 对角线乘积相等



求 R_{AB}



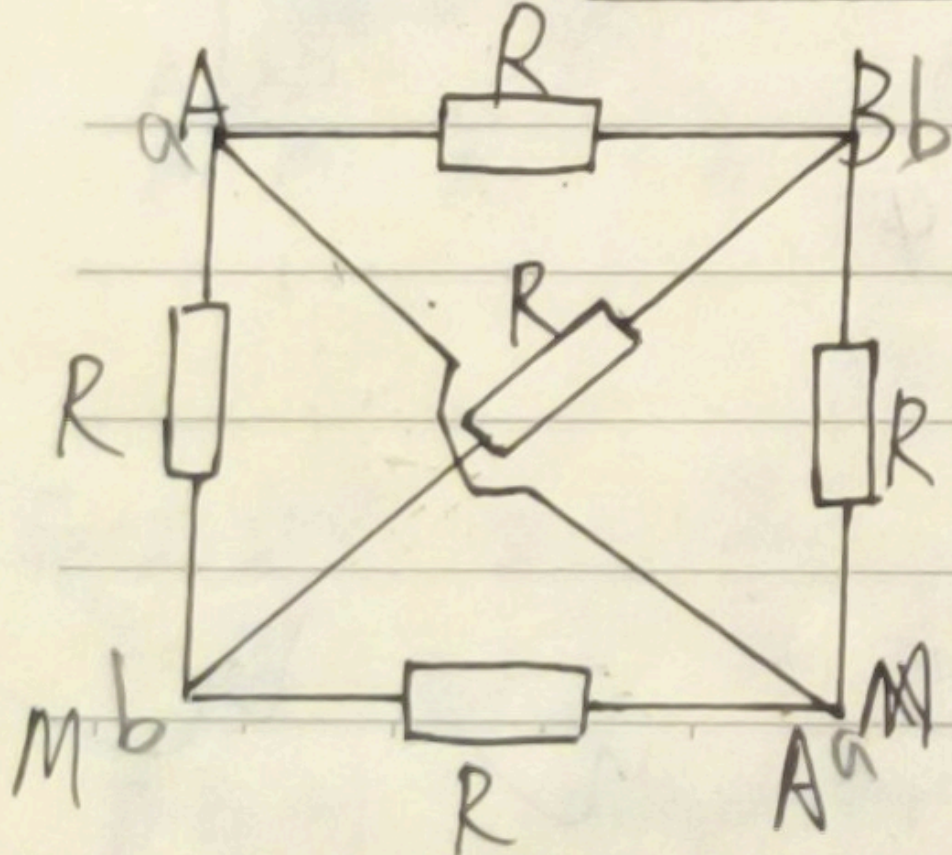
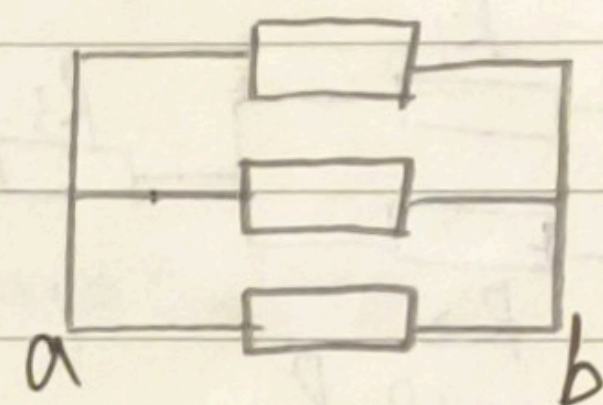
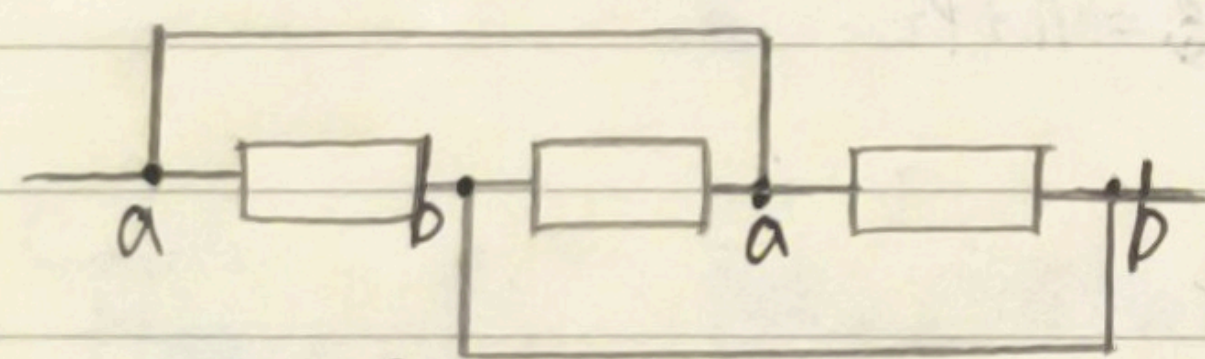
$$R_{AB} = \frac{4 \times 2}{4 + 2} + 1 = \frac{7}{3} \Omega$$

(3) 节点法

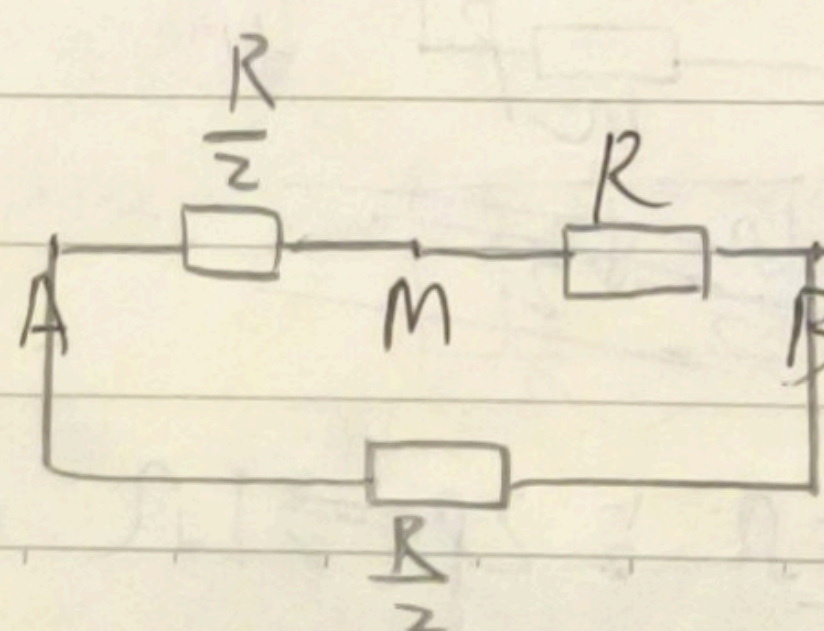
由2根及以上导线汇交而成的点

等势原理：电压 = 电势降

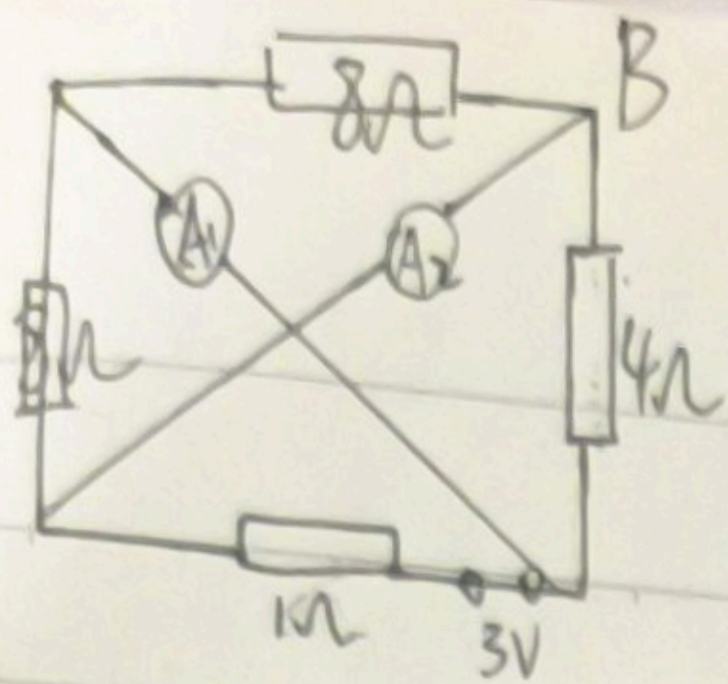
在一根导线上所有点电势相等



求 R_{AB}



$$\frac{\frac{3}{2} \frac{1}{2}}{\frac{3}{2} + \frac{1}{2}} = \frac{3}{8} R$$



求 A_1 A_2
 $0.5A$ $0.75A$

NO. 1.429g

No.

Date.

戴维南原理——独立方程

含容电路

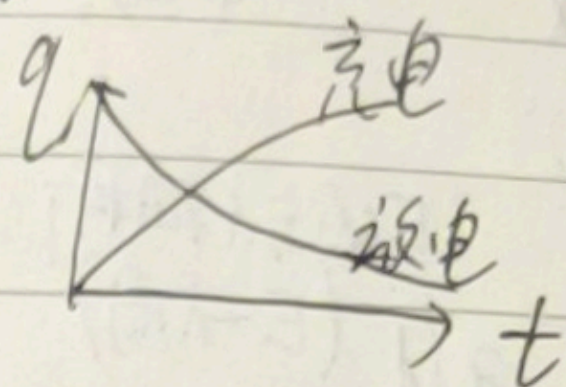
通交隔直

节点 $I_{in} = I_{out}$
 回路 $U = 0$

在一回路有2个方程

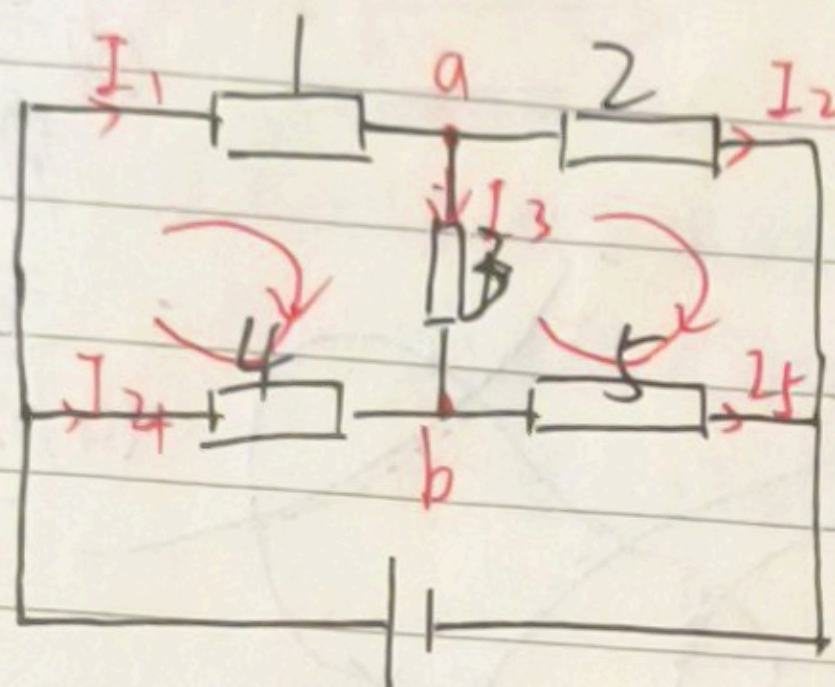
$$\begin{cases} U = IR / E = U + IR \\ P = I^2 R \cdot P = UI \end{cases}$$

与电容器串联的任何用电器(稳定时)不工作



基尔霍夫定律

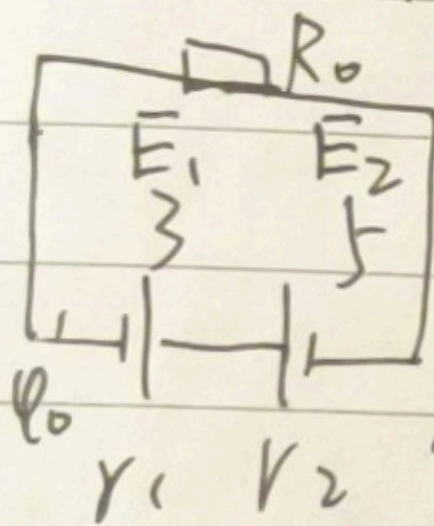
$$\sum I = 0 \quad \sum \varphi = 0$$



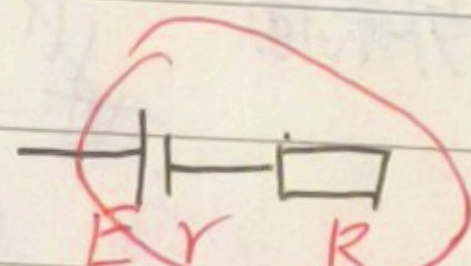
$$\begin{cases} I_1 = I_2 + I_3 \\ I_3 + I_4 = I_5 \\ I_1 + I_4 = I_2 + I_5 \\ \varphi_0 - I_1 R_1 - I_3 R_3 + I_4 R_4 = \varphi_0 \\ \varphi_0 + I_5 R_5 + I_3 R_3 - I_2 R_2 = \varphi_0' \end{cases}$$

戴维南原理——等效电源

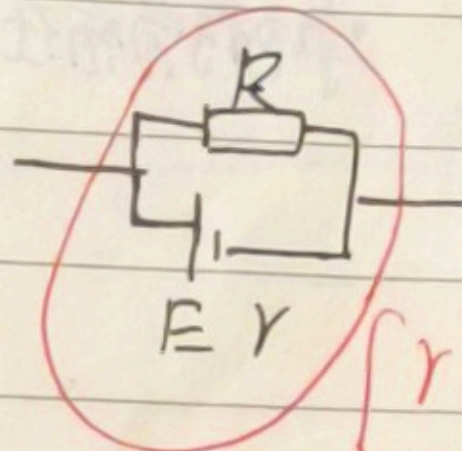
等效电阻: 开路电阻
 等效电压: 开路电压



$$\begin{cases} E = 2V \\ r = r_1 + r_2 \end{cases}$$



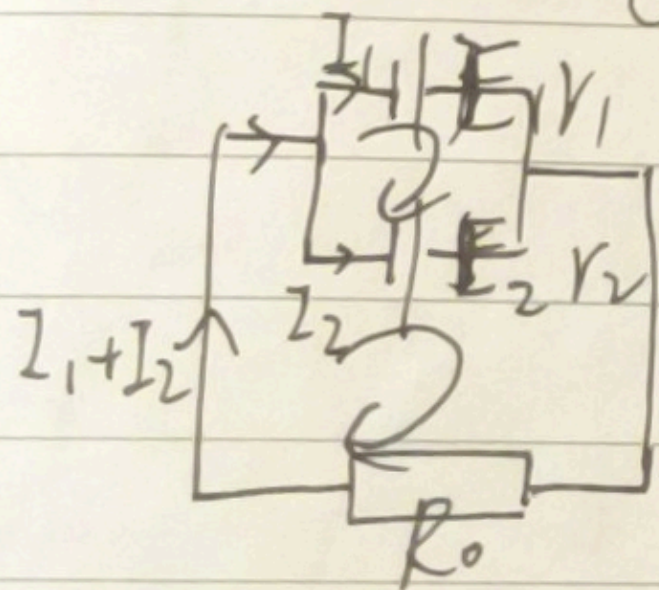
$$\begin{cases} r' = r + R \\ E' = E \end{cases}$$



$$\begin{cases} r' = \frac{Rr}{R+r} \\ E' = U_R = \frac{R}{R+r} E \end{cases}$$

$$\varphi_0 + E_1 - I_1 r_1 - I_2 r_2 + E_2 - I R_0 = \varphi_0$$

$$I = \frac{E_1 - E_2}{r_1 + r_2 + R_0}$$



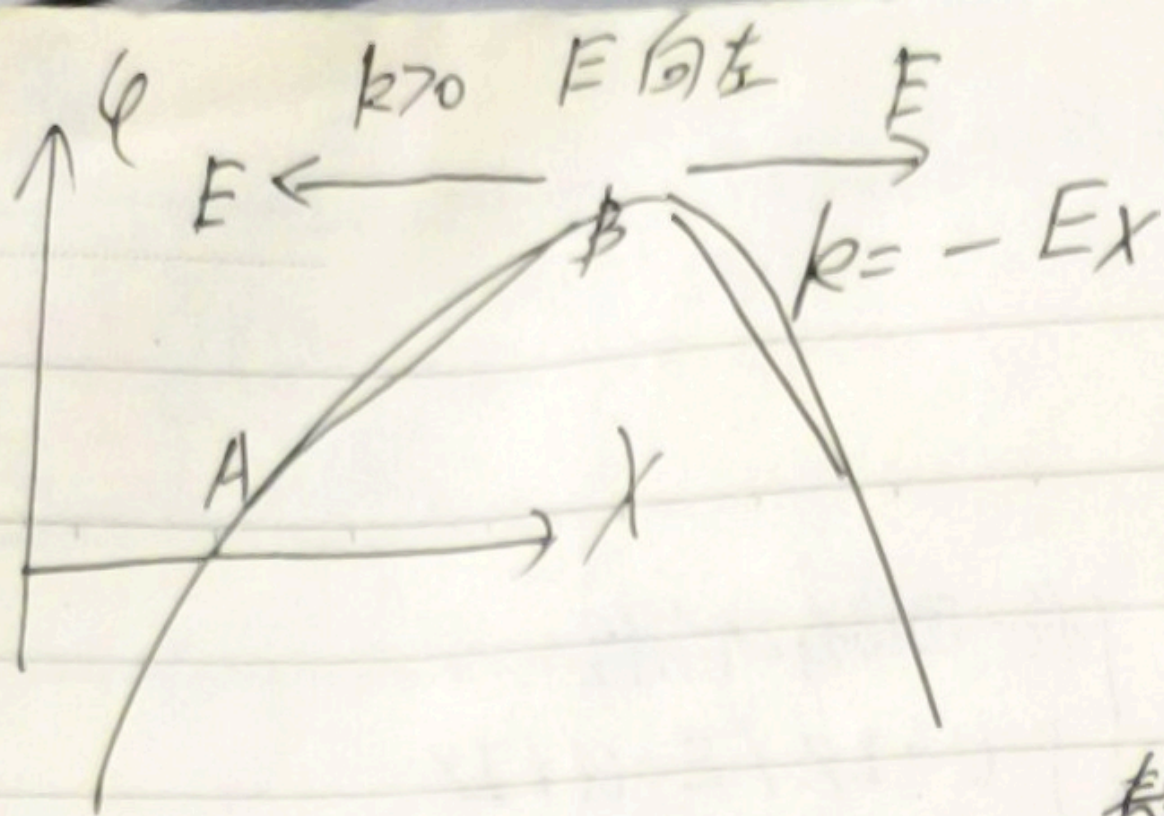
$$\varphi_0 + E_1 - I_1 r_1 - E_2 + I_2 r_2 = \varphi_0$$

$$\varphi_0 + E_2 - I_2 r_2 - (I_1 + I_2) R_0 = \varphi_0$$

$$I = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2 + (r_1 + r_2) R_0} = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2 + (r_1 + r_2) R_0} = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2 + (r_1 + r_2) R_0}$$

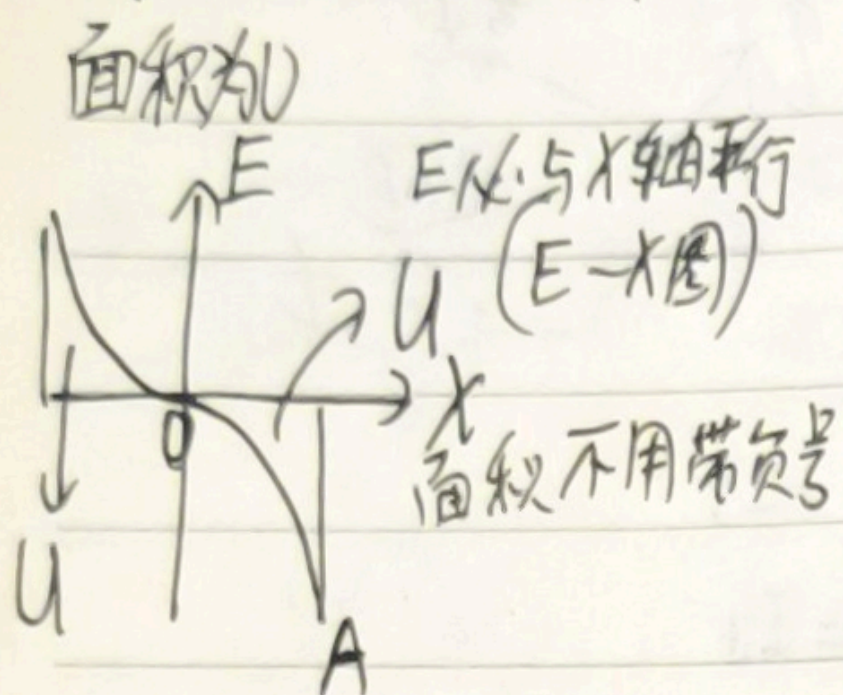
No.

Date.



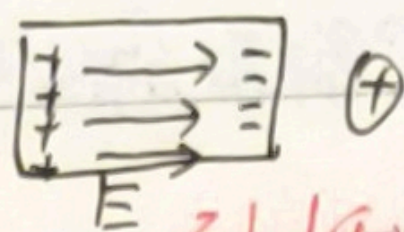
E-X

φ-X



静电感应/平衡/屏蔽

过程 结果 应用

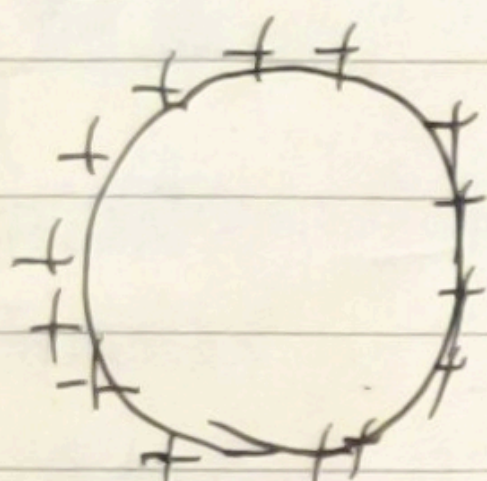
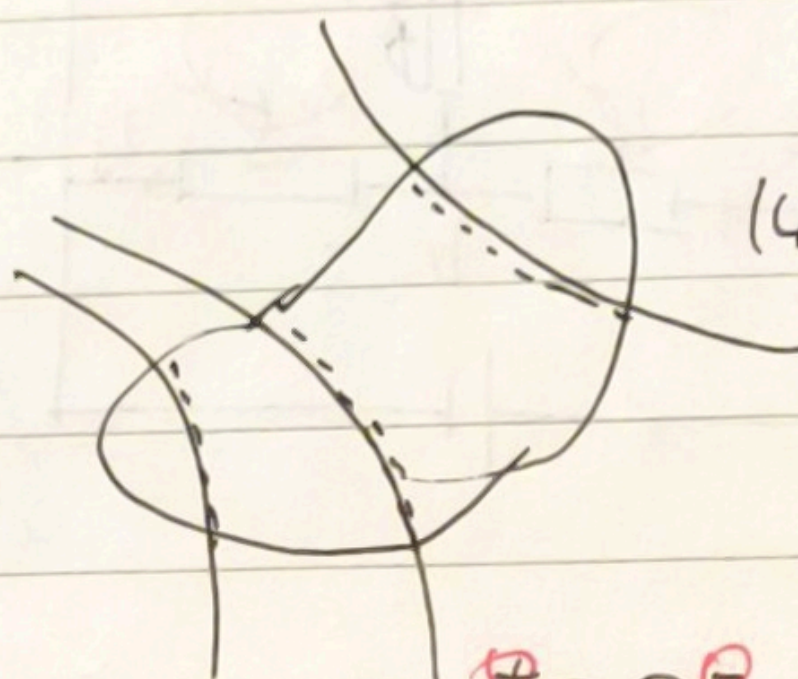
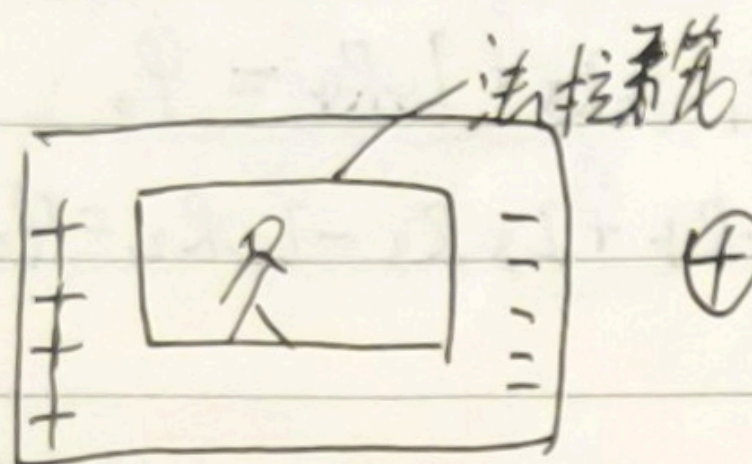


(1) 近异远同

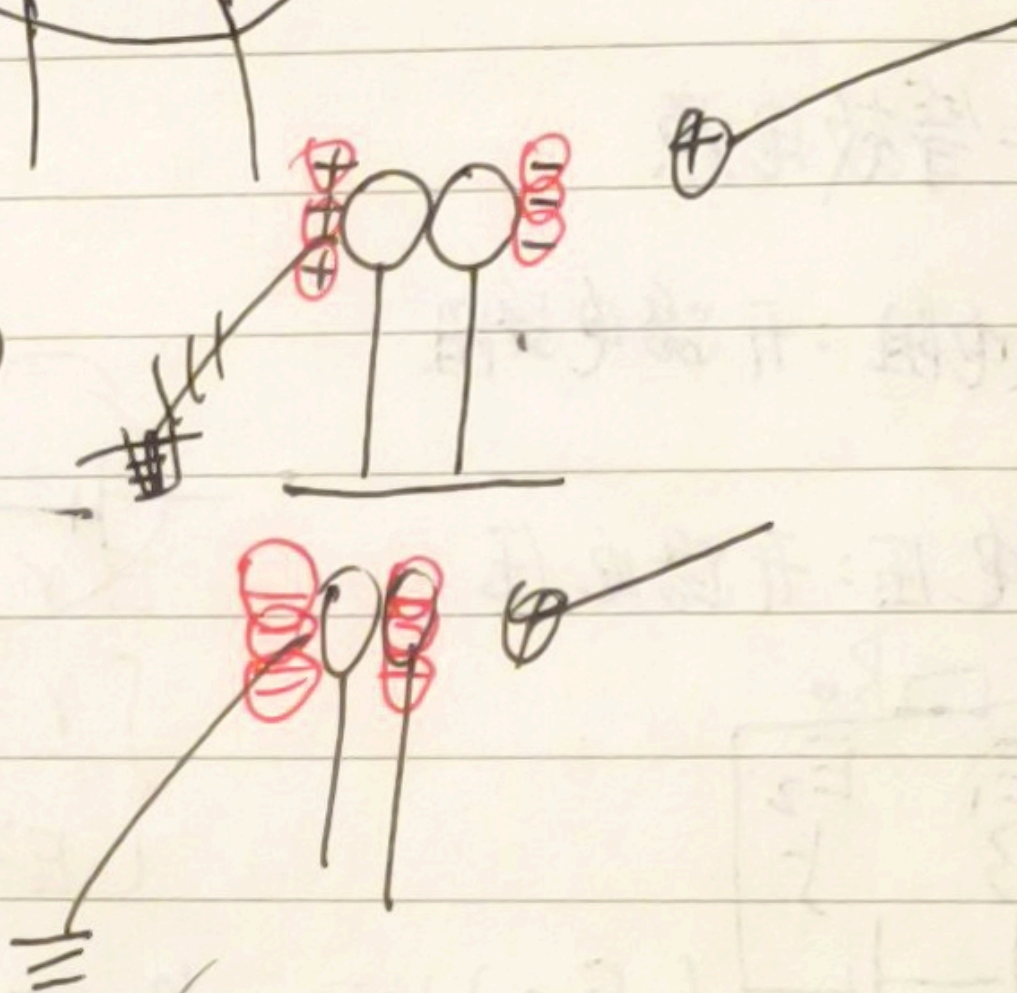
(2) 内部无净电荷场强为0

(3) 导体为等势体——表面为等势面——电场线处处垂直等势面

(4) 内部挖空：静电屏蔽



净电荷分布在导体外表面

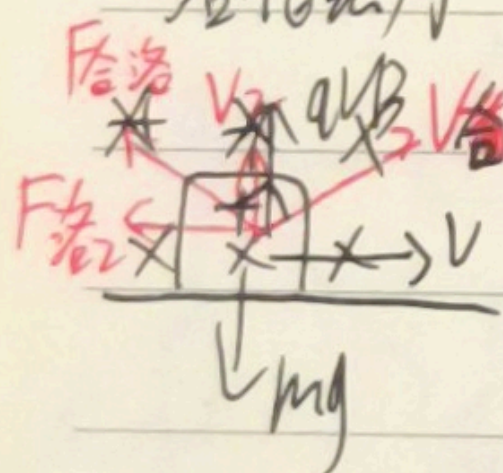


$$B = \frac{F}{IL \sin \theta}$$

电生磁——连——手——右手

洛伦兹力 分力可以做功，合力不做功

使小木块可与地分离求磁场移动速度



$$qvB = mg \quad \text{使 } B \text{ 向左以 } v = \frac{mg}{qB} \text{ 运动}$$